



TEAM NAME: TO CONFIRM

Team members: to confirm

Singapore Polytechnic

BUSGUARD

Connected passenger assistance

DRAFT • team details pending

Inclusive boarding, coordinated assistance

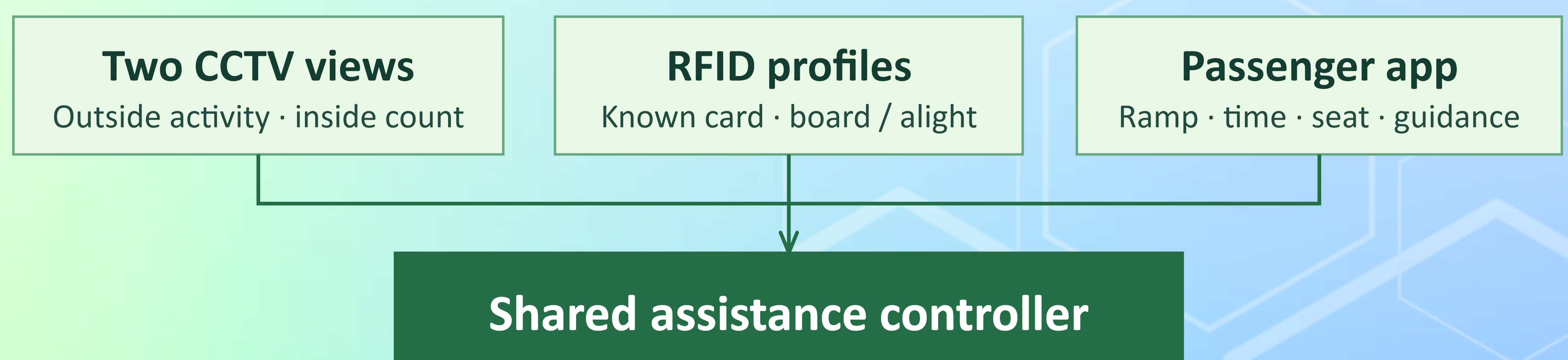
Passengers need a clear way to ask for help and enough time to board.

BusGuard connects two CCTV views, RFID profiles and passenger requests through one local controller.



3D bus visualization • vehicle outputs are simulated

One controller keeps every interface in step



Detection console • 3D bus • passenger updates • clear announcements

Time for the passenger

15 s quiet window. RFID: 10 s from a tap. Ramp request: 20 s. Admission closes at 50 s; accepted time finishes in full.

Fresh evidence before departure

After doors close: 5 s of fresh results with no standing detected. Stale feeds and active holds keep departure held.

Physical inputs: CCTV + ESP32 RFID. Simulated outputs: ramp, doors and bus motion.

